Slicing Keeps the Subobjects (§1.63)

by

§1.63 — $\Sigma : \boldsymbol{A}/B \to \boldsymbol{A}$ identifies every subobject poset of the slice with one of $\boldsymbol{A}$'s, compatibly with inverse images — so "pre-logos" passes to slices for free

The book's statement: for any category $\boldsymbol{A}$ and object $B \in \boldsymbol{A}$, the forgetful functor $\Sigma : \boldsymbol{A}/B \to \boldsymbol{A}$ (send a structure map $X \to B$ to plain X) yields an isomorphism

$$\mathrm{Sub}_{\boldsymbol{A}/B}(-) \cong \mathrm{Sub}_{\boldsymbol{A}}(\Sigma(-)).$$

If $\boldsymbol{A}$ has pullbacks, these isomorphisms respect inverse images. If $\boldsymbol{A}$ has unions, so does $\boldsymbol{A}/B$. Hence if $\boldsymbol{A}$ is a pre-logos so is $\boldsymbol{A}/B$, and $\Delta : \boldsymbol{A} \to \boldsymbol{A}/B$ ($A \mapsto (B \times A \to B)$) is a representation of pre-logoi.

§1. One category, two presentations — one Δ, two readings (§1.544)

$\boldsymbol{A}/B$ has **one definition**: objects are **all** arrows $X \to B$, morphisms are commuting triangles over B. The book presents it twice:

- **(a) plain slice** of $\boldsymbol{A}$ — the form in use from §1.414 on;
- **(b) the §1.544 re-presentation**, in force from §1.544 onward — so also here in §1.63. Build the slice over the **inflation** $\boldsymbol{A}'$ (objects are finite sequences of $\boldsymbol{A}$-objects, the forgetful functor picks a product $A_1 \times \cdots \times A_n$ for each, binary product = **concatenation**), which upgrades product cancellation to strict equality: $B \times A = B \times A' \Rightarrow A = A'$. Then rename the image of Δ to $\boldsymbol{A}$ itself.

The two presentations are equivalent categories; everything true in one is true in the other — the §1.63 iso and the concrete Set/B computation below are the same either way. The gain of (b) is the one Freyd states in §1.544: it lets us "consider $\boldsymbol{A}$ to be a **subcategory** of $\boldsymbol{A}/B$" — the renaming makes Δ injective on objects, so each $\boldsymbol{A}$-object **is** an $\boldsymbol{A}/B$-object, rather than $\boldsymbol{A}$ merely mapping into $\boldsymbol{A}/B$ by a faithful functor. (§1.545–1.546 need this: the capitalization tower is a union of subcategories.) The embedded copy consists of the objects $\Delta(A) = (B \times A \to B)$ — renamed, in (b), to A itself.

Likewise Δ is **one functor** — $A \mapsto (B \times A \to B)$, $f \mapsto 1 \times f$: the diagonal of §1.44, the right half of $(B \times -) = \Sigma \circ \Delta$ — with two readings: in (a) it is a functor between different categories; in (b), after the renaming, the **same** functor is the subcategory inclusion — $\Delta(A)$ **is** A.

Neither presentation shrinks the object class: a general object of $\boldsymbol{A}/B$ is an **arbitrary** arrow $p : X \to B$ — e.g. a proper subobject $B' \hookrightarrow B$ is a slice object whose domain is no product $B \times A$. The objects $B \times A \to B$ are merely the embedded copy of $\boldsymbol{A}$; the slice is strictly larger (that properness is the engine of capitalization). The §1.63 iso quantifies over **all** slice objects — it must, since concluding "$\boldsymbol{A}/B$ is a pre-logos" requires **every** $\mathrm{Sub}_{\boldsymbol{A}/B}(p)$ to be a lattice — which is why §§2–4 below work with a general (X, p). What presentation (b) changes is how two things must be read:

Reading $\boldsymbol{\Delta}$ correctly. Δ is **not** "$X \mapsto (X \to B)$": it is $\Delta(A) = (B \times A \to B)$, with $\Sigma(\Delta(A)) = B \times A$. So on the embedded copy of $\boldsymbol{A}$ the §1.63 iso reads

$$\mathrm{Sub}_{\boldsymbol{A}/B}(\Delta A) \cong \mathrm{Sub}_{\boldsymbol{A}}(B \times A)$$

— subobjects of $B \times A$, **not** of A. Under this identification Δ acts on subobjects by $A' \mapsto B \times A'$, which is exactly $\pi^{\#}$ for the projection $\pi : B \times A \to A$; the claim "Δ is a representation of pre-logoi" therefore says $B \times -$ preserves the lattice operations — $B \times (A_1 \cup A_2) = (B \times A_1) \cup (B \times A_2)$, $B \times 0 = 0$ — an instance of the pre-logos axiom "$f^{\#}$ is a lattice homomorphism" applied to π.

Why the subcategory inclusion matters. §1.63 ends with: any (positive) pre-logos is faithfully representable in a **capital** one. That is the §1.545–1.546 ladder — iterate Δ into a tower $\boldsymbol{A} \subset \boldsymbol{A}^* \subset \boldsymbol{A}^{**} \subset \cdots$ and take its **union**. A union of subcategories only makes sense when each rung really is a subcategory (injective on objects); presentation (b) is what makes it one.

Name clash: two "diagonals". The diagonal **functor** $\Delta : \boldsymbol{A} \to \boldsymbol{A}/B$ is not the diagonal **morphism** $\langle 1, 1\rangle : A \to A \times A$ (the one §1.61 uses). They are related — the generic point that slicing adjoins is $\langle 1, 1\rangle : B \to B \times B$ viewed as a slice map $1 \to \Delta(B)$ — but they are different things. Freyd flags the clash himself in §1.535: Δ is only **conjugate** to the standard diagonal functor under the equivalence $\mathrm{Set}/B \simeq \mathrm{Set}^B$.

Rule.
1. $\boldsymbol{A}/B$ always means the full slice: objects are **all** arrows $X \to B$ — never only the $B \times A \to B$.
2. Δ is the one functor $A \mapsto (B \times A \to B)$. From §1.544 on, read $\Delta(A)$ as A itself, so that $\boldsymbol{A} \subset \boldsymbol{A}/B$ is a subcategory — required wherever $\boldsymbol{A}$ must sit inside $\boldsymbol{A}/B$ (capitalization towers and their unions).
3. On subobjects: $\mathrm{Sub}_{\boldsymbol{A}/B}(\Delta A) = \mathrm{Sub}_{\boldsymbol{A}}(B \times A)$, and Δ acts by $A' \mapsto B \times A'$.
4. Δ (the diagonal **functor**) is not $\langle 1, 1\rangle$ (the diagonal **morphism**).

§2. The two functors — and why the "$(-)$"

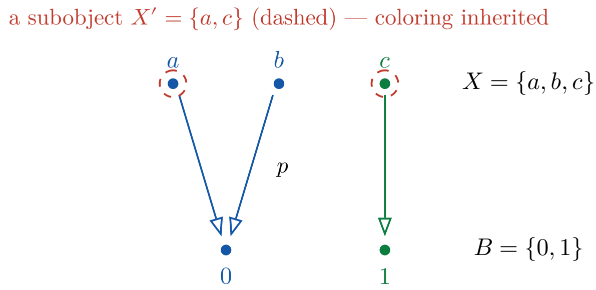
Both sides are **families of posets**, one for each object of the slice:

- $\mathrm{Sub}_{\boldsymbol{A}/B}(-)$: plug in a slice object $p : X \to B$, get the poset of its subobjects **computed in the slice category**;
- $\mathrm{Sub}_{\boldsymbol{A}}(\Sigma(-))$: plug in the **same** slice object, apply Σ (throw away p, keep X), get the poset of subobjects of plain X **computed in $\boldsymbol{A}$**.

Why the placeholder $(-)$ instead of just $\mathrm{Sub}_{\boldsymbol{A}/B}$? Because the isomorphism is between **functors**, and the two subobject functors have different domains: $\mathrm{Sub}_{\boldsymbol{A}/B}$ lives on $\boldsymbol{A}/B$, $\mathrm{Sub}_{\boldsymbol{A}}$ lives on $\boldsymbol{A}$ — "$\mathrm{Sub}_{\boldsymbol{A}/B} \cong \mathrm{Sub}_{\boldsymbol{A}}$" would be ill-typed. The right-hand side is really the **composite** $\mathrm{Sub}_{\boldsymbol{A}} \circ \Sigma$, and $\mathrm{Sub}_{\boldsymbol{A}}(\Sigma(-))$ is how the book writes that composite: the $(-)$ marks where the common argument goes, with Σ inserted on one side only. The notation also matches how the claim unfolds: first an isomorphism of posets **at each object** (no pullbacks needed), then — when $\boldsymbol{A}$ has pullbacks — the next sentence upgrades it to a **natural** isomorphism ("respect inverse images" = naturality in the argument).

§3. The isomorphism, concretely

Take $\boldsymbol{A} = \mathrm{Set}$, $B = \{0, 1\}$. In either presentation an object of Set/B is an arrow into B — a set with a B-coloring: $p : X \to B$ paints every element color 0 or color 1 (presentation (b) additionally embeds SET inside as the objects $B \times A \to B$). Say



What is a subobject of (X, p) in the slice? A monic over B, i.e. a commuting triangle:

$$
\begin{tikzcd}
X' \arrow[r, tail] \arrow[dr, "\text{forced}"'] & X \arrow[d, "p"] \\
& B
\end{tikzcd}
$$

Here is the whole point: the structure map $X' \to B$ has **no freedom** — commutativity forces it to be the restriction of p along the inclusion. So a subobject of (X, p) is **just a subset of $\boldsymbol{X}$** wearing its inherited coloring — e.g. $X' = \{a, c\}$ with a colored 0, c colored 1:

$$\mathrm{Sub}_{\mathrm{Set}/B}((X, p)) = \text{all 8 subsets of } \{a, b, c\} = \mathrm{Sub}_{\mathrm{Set}}(X).$$

Same elements, same inclusion order — that is the isomorphism. (Behind it: a slice morphism is monic in $\boldsymbol{A}/B$ iff its underlying $\boldsymbol{A}$-map is monic, and every mono into X acquires a unique over-B structure.)

What the iso does <u>not</u> say. Hom-sets **do** shrink in the slice: maps $(X, p) \to (Y, q)$ must preserve colors, so there are far fewer of them than maps $X \to Y$. But the **subobjects** do not shrink at all. Slicing changes the maps, not the subobject lattices.

§4. "Respects inverse images" — naturality, concretely

For a slice morphism $f : (X, p) \to (Y, q)$ the claim is that the square

$$
\begin{tikzcd}
\mathrm{Sub}_{\boldsymbol{A}/B}((Y, q)) \arrow[r, "\cong"] \arrow[d, "f^{\#} \text{ in } \boldsymbol{A}/B"] & \mathrm{Sub}_{\boldsymbol{A}}(Y) \arrow[d, "f^{\#} \text{ in } \boldsymbol{A}"'] \\
\mathrm{Sub}_{\boldsymbol{A}/B}((X, p)) \arrow[r, "\cong"] & \mathrm{Sub}_{\boldsymbol{A}}(X)
\end{tikzcd}
$$

commutes. In the example: let $Y = \{u, v\}$ with $q : u \mapsto 0, v \mapsto 1$, and let f be the color-preserving map $a, b \mapsto u$, $c \mapsto v$. Take the subobject $\{v\} \subseteq Y$. Computed in the slice or in $\mathrm{S}\textsc{et}$, $f^{\#}(\{v\}) = \{c\}$ — preimage is preimage, the coloring just tags along. Pullbacks in $\boldsymbol{A}/B$ are computed on underlying objects in $\boldsymbol{A}$, so $f^{\#}$ on the two sides of the iso is **literally the same operation**.

§5. Why Freyd wants this

Every axiom of "pre-logos" is a statement about subobject posets and $f^{\#}$: Sub is a lattice, finite unions exist, $f^{\#}$ is a lattice homomorphism. The iso says the slice has **the same** subobject posets and **the same** $f^{\#}$ as $\boldsymbol{A}$. So all those properties transfer wholesale — $\boldsymbol{A}$ pre-logos $\Rightarrow$ $\boldsymbol{A}/B$ pre-logos, no new proof needed — and the diagonal $\Delta : \boldsymbol{A} \to \boldsymbol{A}/B$, $A \mapsto (B \times A \to B)$, is a representation of pre-logoi (its subobject action $A' \mapsto B \times A'$ is $\pi^{\#}$ — see the §1.544 box above). The same transfer gives the positive case: if $\boldsymbol{A}$ is a positive pre-logos, so is $\boldsymbol{A}/B$.

It also generalizes §1.414, which was the special case "subobjects of the terminator": the terminator of Set/B is id_B, so $\mathrm{Sub}_{\mathrm{Set}/B}(1) = \mathrm{Sub}_{\mathrm{Set}}(B)$ — the four subsets of $\{0, 1\}$ are the **truth values** of Set/B. One reason slices behave like "sets varying over B": slicing moves $\mathrm{Sub}_{\boldsymbol{A}}(B)$ into the terminator position.

⋆ Punchline

Σ is an isomorphism on every subobject poset, commuting with inverse images: slicing rescales the **maps** (color-preserving only), never the **subobject lattices**. Any property expressed purely in $\big(\mathrm{Sub}, f^{\#}\big)$ — lattice, finite unions, distributivity: the pre-logos axioms — therefore holds in $\boldsymbol{A}/B$ the moment it holds in $\boldsymbol{A}$. The $(-)$ in $\mathrm{Sub}_{\boldsymbol{A}/B}(-) \cong \mathrm{Sub}_{\boldsymbol{A}}(\Sigma(-))$ is the placeholder that makes this a statement about **functors** ($\mathrm{Sub}_{\boldsymbol{A}} \circ \Sigma$ on the right), not a single bijection.

§6. §1.631 — complemented subobjects: the complement is unique

§1.63 made every $\mathrm{Sub}_{\boldsymbol{A}}(A)$ a **distributive** lattice, and transported that to slices. §1.631 cashes the distributivity in immediately. Call $A_1 \hookrightarrow A$ a **complemented subobject** if some $A_2 \hookrightarrow A$ satisfies

$$A_1 \cap A_2 = 0 \quad \text{and} \quad A_1 \cup A_2 = A.$$

Freyd's remark: **the distributivity of Sub($\boldsymbol{A}$) makes $\boldsymbol{A_2}$, when it exists, unique.**

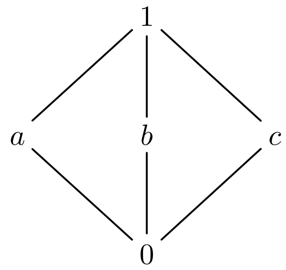
Why distributivity forces it. Suppose A_2 and $A_{2'}$ both complement A_1. Then

$$\begin{aligned} A_2 &= A_2 \cap A = A_2 \cap (A_1 \cup A_{2'}) \\ &= (A_2 \cap A_1) \cup (A_2 \cap A_{2'}) \quad (\star) \\ &= 0 \cup (A_2 \cap A_{2'}) = A_2 \cap A_{2'} \subseteq A_{2'}, \end{aligned}$$

where $(\star)$ is the distributive law $\cap$-over-$\cup$ — the **only** nontrivial step. By symmetry $A_{2'} \subseteq A_2$, so $A_2 = A_{2'}$. Everything rides on that one move.

Unique, concretely. In the Boolean algebra of subsets of $\{1, 2, 3\}$, take $A = \{1, 2, 3\}$ and $A_1 = \{1, 2\}$. A complement needs $A_1 \cap A_2 = \emptyset$ and $A_1 \cup A_2 = \{1, 2, 3\}$: anything containing 1 or 2 overlaps A_1, and anything missing 3 fails to cover — so $A_2 = \{3\}$ is **forced**.

Fails without distributivity. The smallest non-distributive lattice is the diamond M_3:



The three middle elements are pairwise incomparable; each pair meets at 0 and joins at 1. So $a \cap b = 0$ and $a \cup b = 1$ — b complements a — but equally $a \cap c = 0$ and $a \cup c = 1$, so c complements a too. ***a* has two complements.** The break is exactly the distributive step: $a \cap (b \cup c) = a \cap 1 = a$, whereas $(a \cap b) \cup (a \cap c) = 0 \cup 0 = 0$.

No pre-logos looks like M_3. §1.63 (via §1.616) makes Sub(A) **distributive** in every pre-logos — and, by the slice iso above, in every slice too. So the M_3 pathology cannot occur, and complements are always unique. The diamond is precisely the witness that "distributive" is load-bearing in §1.631, not decoration.

⋆ Punchline

Uniqueness of the complement is pure lattice distributivity — no categorical input beyond "Sub(A) is distributive", which §1.63 already transported to every slice. It is formalized as `Freyd.complement_unique` (`Fredy/S1_62.lean`), **axiom-free**: two complements of A_1 are shown mutually $\subseteq$ (hence isomorphic), from the $\subseteq$-half `complement_le_other` applied both ways. M_3 is the witness that the distributivity hypothesis does real work.